Cincinnati Weather Trend and Severe Weather Monitoring Dashboard – Project Proposal

Project Overview

Project Title: Cincinnati Weather Trend and Severe Weather Monitoring Dashboard
Project Type: Interactive Web Application
Target Audience: Cincinnati, OH residents and visitors
Technology Stack: Python, Dash, Open-Meteo API

Background

This project proposes the development of an interactive weather application that provides severe weather alerts and weather trends based on daily weather forecasts for Cincinnati, Ohio. This dashboard will help users better understand temperature trends, precipitation patterns, wind conditions, and potential severe weather events through clear visualizations and alert indicators.

Business Problem

Cincinnati, Ohio experiences frequent weather fluctuations throughout all four seasons, including thunderstorms, heavy rainfall, high winds, flooding, snow, and rapid temperature changes. These changing conditions can make it difficult for residents to monitor weather trends and identify severe weather events efficiently. This project aims to address that issue by developing an interactive weather monitoring dashboard that uses data from the Open-Meteo API, processes the information through a Python-based ETL pipeline, and visualizes weather trends and severe weather indicators in a centralized, user-friendly format.

Objectives

Primary Objectives

- Extract daily weather forecast data from the Open-Meteo API using Python
- Clean and transform JSON weather data into a structured dataset using pandas

- Build a Python-based ETL pipeline to automate data extraction, transformation, and storage
- Develop an interactive Dash dashboard to visualize weather trends and severe weather indicators for Cincinnati

Secondary Objectives

- Identify severe weather conditions using alert thresholds
- Visualize temperature, precipitation, and wind trends over time
- Present weather data in a centralized and easy-to-understand format

Features and Functionality

Core Features

1. **Weather Dashboard**
- Daily forecast summaries
- High and low temperature tracking
- Rainfall and precipitation monitoring
- Wind speed analysis
- Severe weather alert indicators
- Interactive charts and filters
2. **Severe Weather Monitoring**
- Heat Alerts
- Freeze Alerts
- Heavy Precipitation Alerts
- High Wind Alerts
3. **Interactive User Interface**
- Temperature trend line charts
- Precipitation bar charts
- Wind Speed visualizations
- KPI cards for highest temperature, precipitation totals, and severe weather days

Technical Architecture

Backend Components

- **Data Retrieval**: Open-Meteo API integration for weather data

- **Data Processing**: Pandas for data manipulation and transformation
- **Data Storage:** CSV files for processed weather datasets

Frontend Components

- **Visualization**: Dash for interactive charts and graphs
- **Dashboard Features**: Filters, KPI cards, and trend lines

Data Flow

1. **API Integration:** Fetch weather data from Open-Meteo API every day
2. **Data Processing:** Transform raw API data into a user-friendly format
3. **Severe Weather Alert Logic:** Apply severe weather logic to generate alerts
4. **User Interface:** Display weather data, interactive charts, and severe weather alert through interactive dashboard
5. **User Interaction:** Allow filtering and slicers to explore temperature, precipitation, and wind pattern trends

Data Sources and API's

Primary Data Source

- **Primary Data Source:** Open-Meteo API
- **Location:** Cincinnati, OH (39.1271°N, 84.5144°W)
- **Data Parameters:**
    - Daily maximum and minimum temperature
    - Precipitation totals
    - Wind speed
- **Units:**
    - Fahrenheit
    - MPH
    - Inches
- **Timezone**: America/New York

Severe Weather Alert Logic

Temperature:

- Below 32°F → Freeze Alert
- Above 90°F → Heat Alert
- May include additional alerts with more extreme measures

Precipitation:

- Heavy precipitation threshold alerts (over 1.5 inches)
- May include additional alerts with more extreme measures

Wind Speed

- High wind alerts for strong wind conditions (over 35mph)
- May include additional alerts with more extreme measures

Implementation Plan

Phase 1: Foundation (Week 1)

- Research weather APIs, select Open-Meteo API, finalize project proposal, and set up Python development environment

Phase 2: Building API extraction + Transform Pipeline (Week 2)

- Build API extraction scripts, retrieve weather forecast data for Cincinnati, and begin JSON transformation using pandas

Phase 3: Data Clean + Storage (Week 3)

- Complete data cleaning and transformation pipeline, create severe weather alert logic, and store processed data in CSV format

Phase 4: Building Dashboard (Week 4)

- Build dashboard and visualizations in Dash
- Refine severe weather alert logic

Phase 5: Final Testing (Week 5)

- Final testing and debugging
- Prepare project documentation
- Finalize presentation materials

Expected Outcomes

- Successfully extract daily weather forecast data from Open-Meteo API
- Build a functional ETL pipeline that cleans, transforms, and stores weather data for Cincinnati
- Create a structured weather dataset containing temperature, precipitation, wind speed, and severe weather indicators
- Develop an interactive Dash dashboard that visualizes weather trends and forecast patterns
- Implement severe weather alert indicators for conditions such as heavy precipitation, high winds, freezing temperatures, and extreme heat
- Provide users with a centralized and user-friendly platform for monitoring weather conditions and trends

Conclusion

This project demonstrates a complete data engineering workflow by collecting weather forecast data from the Open-Meteo API, transforming the data through a Python ETL pipeline, and visualizing the results in an interactive Dash dashboard. The application will provide users in Cincinnati with a centralized platform for monitoring weather trends and severe weather conditions while demonstrating practical skills in API integration, data processing, and data visualization.